

Tutorial 9: Exercises 1 and 2

Gradient Descent

Exercise 1: Function

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Equivalently,

$$f(x, y) = (x + 1)^2 + y^2 + 3.$$

This form already suggests that the minimum is reached at

$$x = -1, \quad y = 0.$$

Exercise 1: Jacobian / Gradient

Since f is scalar-valued, its Jacobian is the gradient:

$$\nabla f(x, y) = \begin{pmatrix} f_x \\ f_y \end{pmatrix}.$$

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Therefore,

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Exercise 1: Hessian Matrix

The Hessian is

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$$\det(H_f - \lambda I) = 0.$$

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Therefore, the characteristic equation is

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Therefore,

$$\boxed{(x^*, y^*) = (-1, 0)}.$$

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Thus

$$f_{\min} = 3.$$

Exercise 1: Gradient Descent Method

The gradient descent formula is

$$X_{k+1} = X_k - \rho_k \nabla f(X_k).$$

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So

$$X_{k+1} = \begin{pmatrix} x_k \\ y_k \end{pmatrix} - \rho_k \begin{pmatrix} 2x_k + 2 \\ 2y_k \end{pmatrix}.$$

Therefore,

$$x_{k+1} = x_k - \rho_k(2x_k + 2),$$

$$y_{k+1} = y_k - \rho_k(2y_k).$$

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At each iteration, we move in the descent direction:

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The best step ρ is the one that minimizes:

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So instead of minimizing $f(x, y)$ in two variables, we minimize

$$g(\rho)$$

in one variable.

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Thus

$$\boxed{\rho^* = \frac{1}{2}}.$$

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Update:

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Stopping criterion:

$$\|\nabla f(X_k)\| \leq \varepsilon.$$

Exercise 1: Example with $X_0 = (3, 5)$

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So the method reaches the optimum in one iteration.

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We minimize

$$f(x, y) = 4x^2 + 6y^2 + 6xy + 3x + 4y + 6.$$

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- 2 by gradient descent with fixed step $\rho = 0.1$,
- 3 by gradient descent with dynamic step ρ_k .

Exercise 2: Analytical Minimum

The gradient is

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Solving:

$$x^* = -\frac{1}{5}, \quad y^* = -\frac{7}{30}.$$

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Solving:

$$x^* = -\frac{1}{5}, \quad y^* = -\frac{7}{30}.$$

Approximately:

$$x^* \approx -0.2, \quad y^* \approx -0.2333.$$

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$$\det(H_f) = 8 \cdot 12 - 6^2 = 96 - 36 = 60 > 0.$$

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Therefore, H_f is positive definite.

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Hence f is strictly convex.

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Therefore, H_f is positive definite.

Hence f is strictly convex.

So

$$\left(-\frac{1}{5}, -\frac{7}{30} \right)$$

is the global minimum.

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Thus

$$X_1 = \begin{pmatrix} -0.3 \\ -0.4 \end{pmatrix}.$$

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Compute:

$$\nabla f(X_1) = \begin{pmatrix} 8(-0.3) + 6(-0.4) + 3 \\ 6(-0.3) + 12(-0.4) + 4 \end{pmatrix}.$$

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So:

$$X_2 = \begin{pmatrix} -0.3 \\ -0.4 \end{pmatrix} - 0.1 \begin{pmatrix} -1.8 \\ -2.6 \end{pmatrix}.$$

Therefore:

$$\begin{pmatrix} -0.12 \\ -0.12 \end{pmatrix}$$

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The optimal step is obtained by solving:

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So again, we reduce the problem to a one-variable minimization.

Exercise 2: Dynamic Step — Iteration 1

Start from:

$$x_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

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We have:

$$\nabla f(X_0) = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

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Thus:

$$X(\rho) = X_0 - \rho \nabla f(X_0) = \begin{pmatrix} -3\rho \\ -4\rho \end{pmatrix}.$$

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After substitution:

$$g(\rho) = 204\rho^2 - 25\rho + 6.$$

Exercise 2: Dynamic Step — First Step

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So:

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Thus:

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Solve:

$$g'(\rho) = 0.$$

So:

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Hence:

$$\rho_0 = \frac{25}{408} \approx 0.0613.$$

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Solve:

$$g'(\rho) = 0.$$

So:

$$408\rho - 25 = 0.$$

Hence:

$$\rho_0 = \frac{25}{408} \approx 0.0613.$$

Then:

$$X_1 = X_0 - \rho_0 \nabla f(X_0).$$

Exercise 2: Dynamic Step — X_1

We have:

$$X_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \frac{25}{408} \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

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$$X_1 = \begin{pmatrix} -\frac{25}{136} \\ -\frac{25}{102} \end{pmatrix}.$$

Approximately:

$$X_1 \approx \begin{pmatrix} -0.1838 \\ -0.2451 \end{pmatrix}.$$

Exercise 2: Dynamic Step — Iteration 2

Now:

$$x_1 = \begin{pmatrix} -\frac{25}{136} \\ -\frac{25}{102} \end{pmatrix}.$$

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Compute:

$$\nabla f(X_1) = \begin{pmatrix} \frac{1}{17} \\ -\frac{3}{68} \end{pmatrix}.$$

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$$X(\rho) = \begin{pmatrix} -\frac{25}{136} - \frac{\rho}{17} \\ -\frac{25}{102} + \frac{3\rho}{68} \end{pmatrix}.$$

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Define:

$$g(\rho) = f(X(\rho)).$$

Exercise 2: Dynamic Step — Second Step

After substitution, we get:

$$g(\rho) = \frac{23}{2312}\rho^2 - \frac{25}{4624}\rho + \frac{4271}{816}.$$

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Solve:

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Solve:

$$g'(\rho) = 0.$$

So:

$$\frac{23}{1156}\rho = \frac{25}{4624}.$$

Hence:

$$\rho_1 = \frac{25}{92} \approx 0.2717.$$

Exercise 2: Dynamic Step — X_2

Now:

$$X_2 = X_1 - \rho_1 \nabla f(X_1).$$

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Therefore:

$$X_2 = \begin{pmatrix} -\frac{625}{3128} \\ -\frac{4375}{18768} \end{pmatrix}.$$

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Therefore:

$$X_2 = \begin{pmatrix} -\frac{625}{3128} \\ -\frac{4375}{18768} \end{pmatrix}.$$

Approximately:

$$X_2 \approx \begin{pmatrix} -0.1998 \\ -0.2331 \end{pmatrix}.$$

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Approximately:

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This is very close to:

$$X^* = \begin{pmatrix} -0.2 \\ -0.2333 \end{pmatrix}.$$

Exercise 2: Final Results

Analytical minimum:

$$X^* = \begin{pmatrix} -\frac{1}{5} \\ -\frac{7}{30} \end{pmatrix}$$

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Fixed step $\rho = 0.1$:

$$X_1 = \begin{pmatrix} -0.3 \\ -0.4 \end{pmatrix}, \quad X_2 = \begin{pmatrix} -0.12 \\ -0.14 \end{pmatrix}.$$

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Dynamic step:

$$\rho_0 = \frac{25}{408}, \quad \rho_1 = \frac{25}{92}.$$

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Dynamic step:

$$\rho_0 = \frac{25}{408}, \quad \rho_1 = \frac{25}{92}.$$

$$X_2 \approx \begin{pmatrix} -0.1998 \\ -0.2331 \end{pmatrix}.$$